

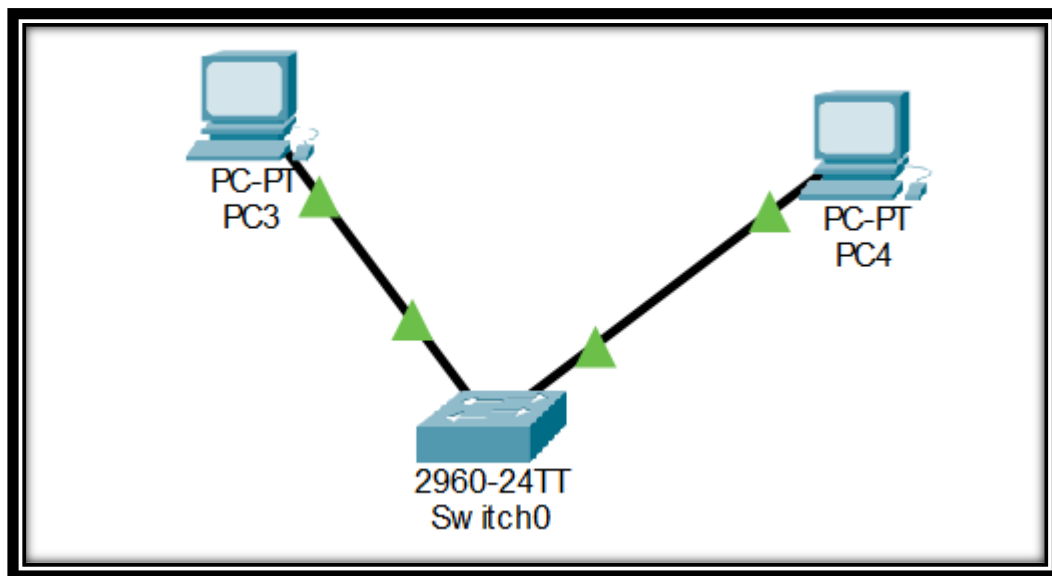


Computer Networks

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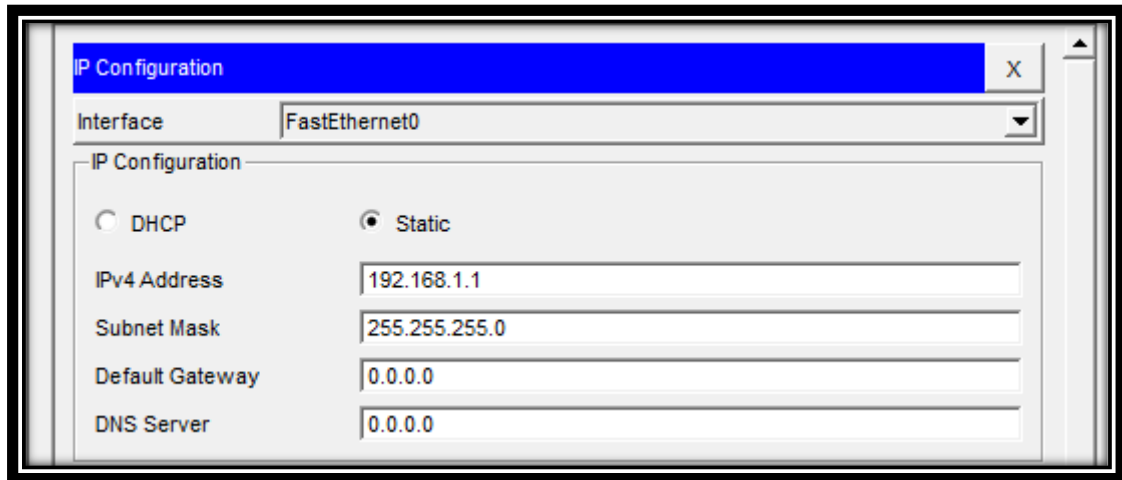
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Network Topology



I created a simple Local Area Network (LAN) topology using Cisco Packet Tracer. I added two end devices (PCs) to the workspace and connected them to a central node, a switch, using copper straight-through cables. The green link lights indicate that the physical connections are active and the devices are ready to communicate within the same network.

IP Configuration



To enable communication, a unique logical IP address was manually assigned to each PC. PC3 was configured with the static IP 192.168.1.1 and PC4 with 192.168.1.2 on the 192.168.1.0 network using a 255.255.255.0 subnet mask. A default gateway was not required because the devices are on the same local network, and no router is used to route traffic externally.

CLI Testing – PC 3

USING Command.. ipconfig

```
FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::202:4AFF:FE5E:109C
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

The output successfully verified that the static IP address and subnet mask entered in the graphical interface were properly applied to the PC's network adapter.

USING Command.. ping

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

The ping command was used to test network connectivity between PC3 and PC4. Receiving four consecutive "Reply from..." messages confirms that data packets are successfully being exchanged between the two nodes across the switch without any packet loss.

USING Command.. tracert

```
C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.2

Trace complete.
```

The tracert command was run to display the specific path the packets took to reach the destination IP address. The output displays only a single hop because both PCs are directly connected to the same local switch, meaning the data does not need to traverse multiple routers to reach its target.

USING command.. arp -a, netstat

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.2           000d.bd79.6105        dynamic
```

```

C:\>netstat

Active Connections

   Proto  Local Address           Foreign Address         State
C:\>

```

The arp -a command successfully displayed the device's Address Resolution Protocol (ARP) table, which maps the known logical IP addresses to their corresponding physical MAC addresses. Additionally, the netstat command was executed to display active network connections and statistics for the system.

CLI Testing – PC 4

Using command.. ipconfig

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20D:BDFF:FE79:6105
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 192.168.1.2
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: | ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                0.0.0.0

```

The output successfully verified that the static IP address and subnet mask entered in the graphical interface were properly applied to the PC's network adapter.

USING command.. ping

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The ping command was used to test network connectivity between PC3 and PC4. Receiving four consecutive "Reply from..." messages confirms that data packets are successfully being exchanged between the two nodes across the switch without any packet loss.

Using command.. tracert

```
C:\>tracert 192.168.1.1

Tracing route to 192.168.1.1 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1

Trace complete.
```

The tracert command was run to display the specific path the packets took to reach the destination IP address. The output displays only a single hop because both PCs are directly connected to the same local switch, meaning the data does not need to traverse multiple routers to reach its target.

USING command.. arp -a, netstat

```
C:\>arp -a

Internet Address      Physical Address      Type
192.168.1.1           0002.4a5e.109c        dynamic
```

```
C:\>netstat

Active Connections

Proto Local Address          Foreign Address         State
C:\>
```

The arp -a command successfully displayed the device's Address Resolution Protocol (ARP) table, which maps the known logical IP addresses to their corresponding physical MAC addresses. Additionally, the netstat command was executed to display active network connections and statistics for the system.

